

The minimal SIR model, validated then extended to social networks and stochastic differential equations

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Abstract

A discrete stochastic process which roughly follows the mantra of exponential growth that was repeated *ad nauseam* during Covid-19 is presented firstly as a child's game of tag and then as a game of cards. When the results of multiple games are averaged the resulting difference equations match the differential equations of the standard SIR model and thus this minimal model is validated. Extending to an animal social network formed by voles allows comparison of results with those of the complete graph of the basic game. This informs the understanding of flattening the curve and of seasonality. The discrete probability distribution is identified as the Hypergeometric which, lacking a continuous analogue, leads to a discussion of the limits of stochastic differential equation models.

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1. Introduction

Do you know how an epidemic grows? According to Malthus [1], an early economist, populations grow exponentially whilst resources grow linearly. There is no explanation of what happens and when to cause either finite variable to leave their curve to infinity. Applying Occam's Razor leads to the conclusion that Malthus was wrong on both populations and resources.

The public health statistician Farr [2] fitted both quadratics and cubics to the logarithm of infections rather than the trivial task of drawing the straight line for exponential growth. Completing the square on the former leads to a Normal distribution that has since become known as Farr's Law even though the curve is symmetrical and the data is skewed.

In his seminal work Darwin [3] explicitly called out Malthus and debunked his ideas using a *Reductio ad absurdum* argument:

It is the doctrine of Malthus applied with manifold force to the whole animal and vegetable kingdoms; for in this case there can be no artificial increase of food, and no prudential restraint from marriage. Although some species may be now increasing, more or less rapidly, in numbers, all cannot do so, for the world would not hold them.

There is no exception to the rule that every organic being naturally increases at so high a rate, that if not destroyed, the earth would soon be covered by the progeny of a single pair. Even slow-breeding man

has doubled in twenty-five years, and at this rate, in a few thousand years, there would literally not be standing room for his progeny.

During Covid-19 members of the general public in the UK repeated Darwin's argument by asking when the number of infected would exceed the population size. It is tempting to ascribe the popularity of exponential growth in the UK media to the propensity of the British people to brag about how bad they are at Maths. Even du Sautoy [4], the Mathematician and communicator, explained to the layperson how epidemics grow exponentially and he was not alone among the Mathematical community. The public health community was much worse either through a lack of education or a desire to scare.

This viewpoint doesn't seem limited to merely Britain, in America Pinker [5] admonished fellow scientists for telling people not to panic over 'exponential growth'. Apart from epidemics he gives two other examples - compound interest and economic growth. Compound interest, dating back to at least the Babylonians, is defined as exponential growth, the cynic might say that more money can always be printed. In finance there is the salutary tale of the eponymous model for the pricing of European options by Black, Scholes and Merton. Their model wrongly assumes geometric Brownian motion wherein the logarithm of asset prices is normally distributed with the mean and variance increasing linearly with time. i.e. The asset price follows a noisy exponential curve. In 1997 Scholes and Merton were awarded the Nobel prize in economics, Black having died in the interim. At the time the two had left academia to work for Long Term Capital Management, the following year the US Federal Reserve intervened to save the market after the use of the model by their employer had racked up billions of dollars in losses. Other examples of exponential growth include grains of rice on a chessboard and pyramid schemes which leads to the question of whether it can occur naturally.

In what has subsequently become known as the standard SIR model Kermack and McKendrick [6] described the infection curve as arising due to competition between logistic growth and exponential decay. This is an example of a compartmental model, as introduced by Ross and Hudson. A constant population of size N is split into the compartments S (susceptible to infection), I (infected and infectious) and R (recovered and immune to re-infection).

$$N = S + I + R \quad (1)$$

Individuals progress from one compartment to the next as shown in figure 1, hence minus S and plus R are both monotonically increasing with time.

Differentiating equation 1 with respect to time, t , and re-arranging results in an equation where the reproduction rate, $-dS/dR$, has been replaced without loss of generality by the product of its maximum value, R_0 , and some proportion X .

$$\frac{dI}{dt} = \left(-\frac{dS}{dR} - 1 \right) \frac{dR}{dt} \equiv (R_0 X - 1) \frac{dR}{dt}, \quad X \in [0, 1] \quad (2)$$

Kermack and McKendrick closed the model by assuming that a cycle of infection takes place over a constant time interval of $1/\gamma$ resulting in exponential decay and that X , now the ratio of growth to exponential growth, is the number of susceptible divided by the total population size resulting in logistic growth.

$$\frac{dI}{dt} = \left(R_0 \frac{S}{N} - 1 \right) \gamma I \quad (3)$$

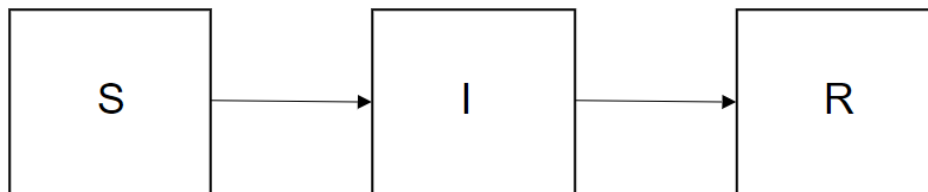


Figure 1. The compartmental model

The SIR model has been extended in numerous ways: More compartments. Fewer compartments. Discrete stochastic behaviour. Geographic grids. Fractional derivatives. Stochastic differential equations. Sparse graphs. Alternative formulae for the reproduction rate. It would be interesting to compare the number of models with the number of datasets.

The mantra of exponential growth as an explanation for epidemics remains alluring because no simple model directly addresses it. Thinking of an epidemic as a game of tag then the game becomes increasingly frenetic as the tagged hunt down the never tagged. It is much easier to believe that the tagging just happens at random. Replace the players with a computer representation then choosing who gets infected requires, for example, the Fisher-Yates [7] shuffle and thus the game of tag begets a game of cards.

2. Agent-Based Model - Epidemic, the card game

For ease of reproducibility by the reader, the agent-based model will be described as a card game. Take two packs of playing cards and reduce them to the same 31 cards, for example aces to eights minus the ace of spades. One pack will be used for tracking which compartment an individual is in, the other will be used to simulate the mixing of an infected individual. Take the tracking pack and spread it out face up. Shuffle and deal one card, then move the corresponding card, which represents patient zero, from the tracking pack into a column of its own. To form a new column, consider each card in the last formed column individually. Remove the corresponding card from the mixing pack, shuffle and deal two cards. If either of those two cards hasn't been dealt before then they can be infected so move their corresponding card from the tracking pack to the new column before returning the three cards back to the mixing pack. Repeat to create more columns until the new column has no cards in it.

This card game implements an SIR model in which the columns of the cards are the generations of the epidemic and R_0 equals two. The susceptible are the remains of the tracking pack. The infected/infectious are contained in the column that is currently being traversed. The recovered are the columns of the older generations. At the end of the game, the columns represent those with acquired immunity and the remains of the tracking pack represent those who have been protected by herd immunity. Herd immunity is seen to arise from the infected mixing with those who aren't susceptible. The reproduction rate may be calculated by dividing the size of a column by the size of the previous column.

$$S_{n+1} = S_n - I_{n+1} \quad (4)$$

$$R_{n+1} = R_n + I_n \quad (5)$$

$$\frac{S_{n+1} - S_n}{R_{n+1} - R_n} = \frac{I_{n+1}}{I_n} \quad (6)$$

The odds of the game matching exponential growth may be calculated in a similar manner to the solution of the birthday problem. The probability decreases from certainty for the first three cards to about evens after 8 cards before shrinking to about 2 in a trillion at the last card.

$$\frac{31}{31} \times \frac{30}{30} \times \frac{29}{29} \times \frac{28}{30} \times \frac{27}{29} \times \cdots \times \frac{3}{29} \times \frac{2}{30} \times \frac{1}{29} = \frac{31!}{31 \times 30^{15} \times 29^{15}} \quad (7)$$

Games of 1000 individuals were played 100 times and the results averaged. This was repeated for four different values of $R_0 - 2, 3, 4$ and 5 . The curves for the average number of infected and the recovered when $R_0 = 2$ are shown in figures 2 and 3, the latter illustrates the effect of herd immunity. $X = 1$ globally refers to the classic exponential curve, whereas $X = 1$ locally refers to the value calculated by assuming exponential growth from the previous average. Obviously, $X = 0$ is the horizontal axis for figure 2.

Central differences for S and R were computed from the averaged values, as was the corresponding interpolation of S at the mid-points between generations. A simple plot produced the same relationship as the standard SIR model as shown in figure 4. Further studies were carried out with different initial conditions and led to the same formula.

$$\frac{S_{n+1} - S_n}{R_{n+1} - R_n} = R_0 \frac{S_{n+1} + S_n}{2N} \quad (8)$$

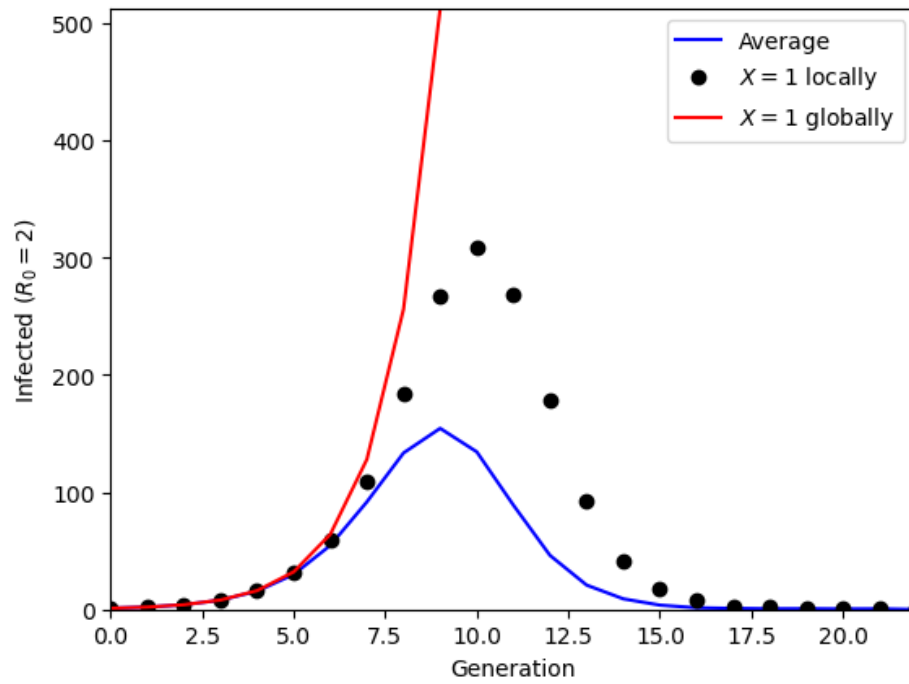


Figure 2. Infected for 1000 individuals

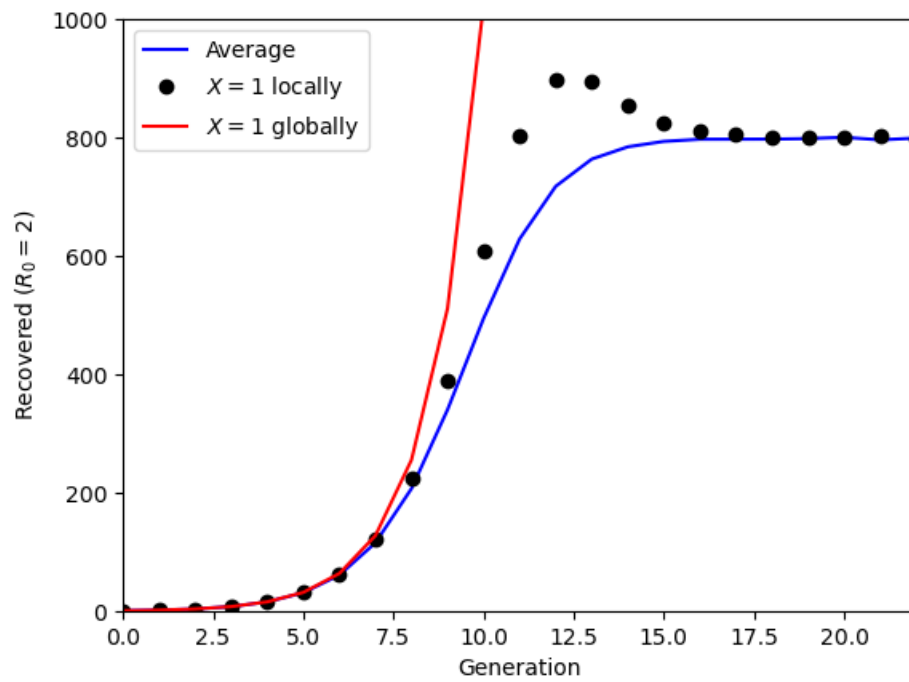


Figure 3. Recovered for 1000 individuals

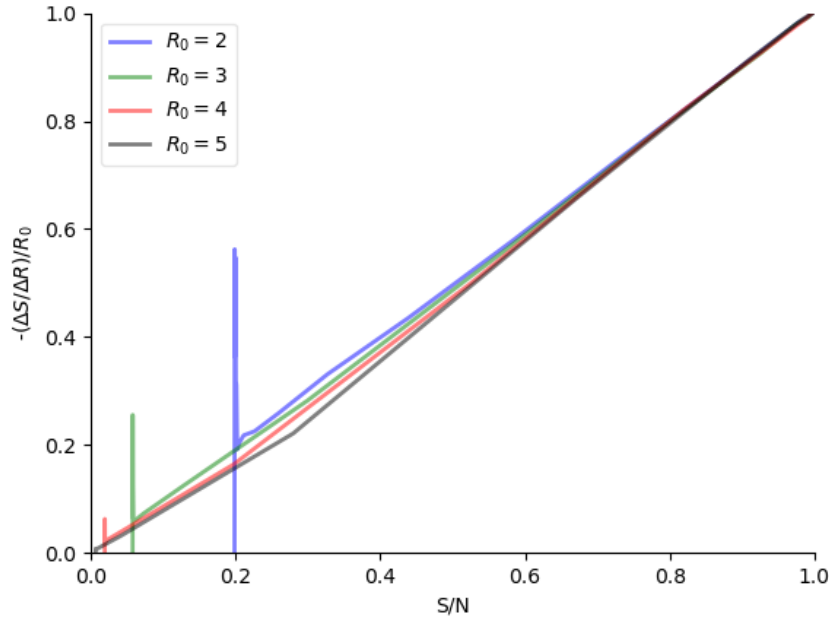


Figure 4. Correlation for the average of multiple games of 1000 individuals

3. Variations on the rules

There are several basic variations that can be made to the game: Model prior immunity by removing cards from the tracking pack. Not returning the dealt cards to the mixing pack results in exponential growth. Dealing from the mixing pack for all cards in the columns results in an SI model. Introducing a third pack allows spectators to gamble on the game. If they draw patient zero then they could win the wooden spoon however, to keep an audience engaged for longer, they could bet on whether their card never ends up infected due to herd immunity. From the standard model the epidemic ends with 5.97 cards having never been infected. Thus, house odds for the dealer to offer would be eight to one leaving the gambler with an expected return of 92 pence on every pound wagered.

Agent-based models are usually heterogeneous however the card game is, with the exception of chance, homogeneous. Since each of the 30 cards has an equal chance of being dealt the population can be described as well mixed and the underlying set of connections between the individuals forms a complete graph. It is fairly trivial to extend the code to instead work on a sparse graph by effectively dealing from a reduced pack corresponding to the contacts of the infected. This was done for an animal social network of 1610 nodes obtained as the largest subgraph from a study of voles, Davis et al. [8] (the data archived by Rossi and Ahmed [9]), in which the degree of the nodes varied from 1 to 37. Each and every node had patient zero assigned to it and the calculations were repeated 10 times with R_0 equal to 2. The average results compared to a complete graph (standard SIR model) of the same size can be seen in figure 5.

The results may be interpreted as the complete graph representing normal behaviour and the social network representing government action to flatten the curve. After a 105 generations 892 voles had never been infected compared to 25 generations and 323 individuals. Hence, in the absence of other measures, flattening the curve drags out an epidemic and results in further waves due to a lack of acquired immunity.

Another interpretation has an influenza-like disease with the social network representing a clement summer when people mix outside and open windows whereas the complete graph represents winter. The peaks of the graphs correspond to the herd immunity threshold, for the voles this is when 180 have been infected whereas, for the complete graph, 862 have been infected thus showing seasonality.

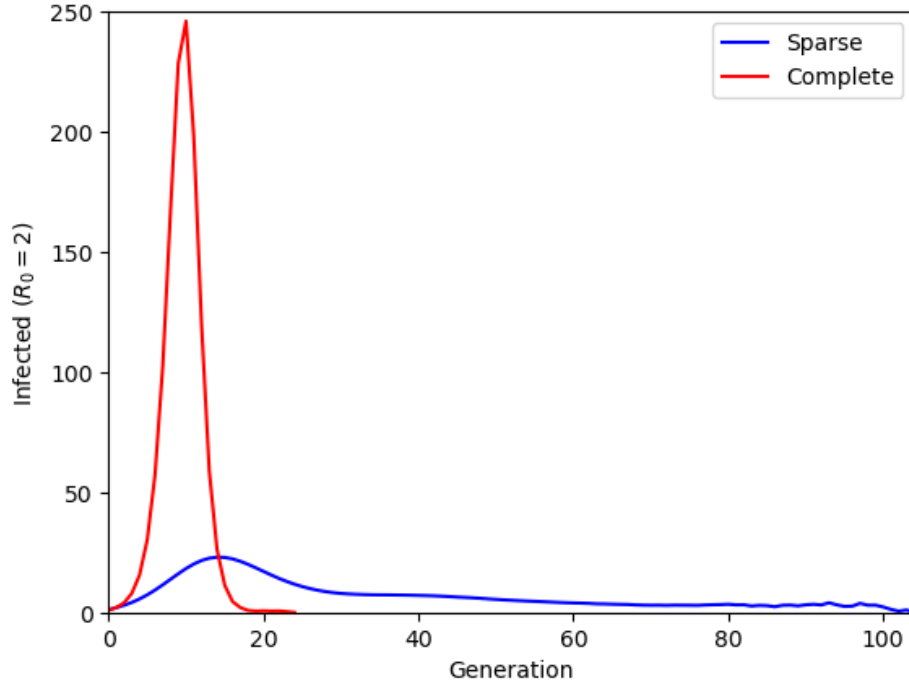


Figure 5. Flattening of the curve due to a sparse graph

4. Stochastic differential equations

The method of choosing who gets infected is known as selection without replacement. The corresponding probability distribution is the Hypergeometric. For a finite population of size N where there are n draws with K potential successes and x actual successes the probability mass function is given in terms of the binomial coefficients as

$$f(x; N, K, n) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}} \quad (9)$$

The distribution has the mean

$$\mu = n \frac{K}{N} \quad (10)$$

Defining the chance of success as $p = K/N$ then the mean is seen to have the same value as that of the Binomial distribution which is the corresponding distribution for selection with replacement. The variance is

$$\sigma^2 = n \cdot \frac{K}{N} \cdot \left(1 - \frac{K}{N}\right) \cdot \frac{N-n}{N-1} \quad (11)$$

This distribution has the property that n draws followed by m draws is the same as $n + m$ draws. Hence, in terms of the epidemic model, each generation involves $R_0 I$ draws resulting in a mean of

$$\mu \equiv \mathbb{E}[X] = R_0 I \frac{S}{N} \quad (12)$$

which, of course, appears in the standard SIR model as the logistic growth term. The variance is

$$\sigma^2 = R_0 I \cdot \frac{S}{N} \cdot \left(1 - \frac{S}{N}\right) \cdot \frac{N - R_0 I}{N - 1} \quad (13)$$

Plain additive noise	σ is constant
Scaled additive noise	$\sigma = \sqrt{\gamma R_0 IS / N}$
External parameter noise	$\sigma \propto \gamma R_0 IS / N$
Heterogeneous parameter noise	$\sigma \propto \gamma R_0 \sqrt{IS} / N$

Table 1. Textbook noise

Stochastic simulations typically use a random number generator to create a value from a Uniform distribution and then use an inverse cumulative distribution function to acquire the value of the random variable. In the case of the Hypergeometric distribution this is expensive and typically the Normal distribution with the same first two moments is used instead even though this cannot reproduce the asymmetry of the Hypergeometric distribution. Note that the Normal distribution is the continuous analogue of the Binomial.

In their textbook Keeling and Rohani [10] detail four stochastic extensions to the standard SIR model as repeated in table 1. These all have the same mean but use different variances with the Wiener process dW_t . Note that by using such a process the value calculated for the reproduction rate now needs to be clamped as it can technically vary from minus to plus infinity.

$$dI = \left(R_0 \frac{S}{N} - 1\right) \gamma I dt + \sigma dW_t \quad (14)$$

None of these proposed noises match that of selection without replacement as given in equation 13. Nor do they match the variance of selection with replacement as given by the Binomial distribution,

$$\sigma^2 = R_0 I \cdot \frac{S}{N} \left(1 - \frac{S}{N}\right) \quad (15)$$

Given the descriptions in the textbook some of these approaches seem to be equivalent to fuzzing rather than modelling the observed noise. i.e. Attempting to find the limits of stability of the system. Note that Normal distributions can be added together to produce another Normal distribution whose mean is the sum of the means and whose variance is the sum of the variances so long as they are independent.

5. Discussion

There are stochastic models which do not have an expected value that matches the standard SIR model. An example being the first discrete stochastic SIR model, the chain binomial model of Reed-Frost [11]. It assumes that each individual contacts every other and that a probability of transmission governs the spread. Although initially used as a teaching model, and demonstrated using marbles, it is not considered a toy model and there is a strand of model development based upon it. It results in a different rate to logistic growth so the minimal model may supersede it.

Science is about the best model winning, hopefully this work inspires a culling of the less fit models from the current surfeit. It should not be possible to get a new model published that doesn't address how it was validated against more than one dataset. The advantage of agent-based models is that they can be simplified to reproduce the same results as classical models and then complicated to, hopefully, reproduce real world behaviour.

The games of tag and cards should be used to educate the general population, particularly children, about the growth of epidemics. Do you know how an epidemic grows? One infects two, two infect four or three or two or one or none. All with different probabilities and a weighted average of logistic growth.

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